Mathematics - I

UNIT - V DIFFERENTIAL EQUATIONS (Higher order)

Two Marks

1. Find the particular integral of $\frac{d^2y}{dx^2} + 2y = x$. [Jan 2014]

Soln:

$$P.I = \frac{1}{2+D^2} x$$

$$= \frac{1}{2\left[1+\frac{D^2}{2}\right]} x$$

$$= \frac{1}{2}\left[1+\frac{D^2}{2}\right]^{-1} x$$

$$= \frac{1}{2}\left[1-\frac{D^2}{2}+\frac{D^4}{4} - \dots\right] x$$

$$= \frac{1}{2}\left[x - \frac{1}{2}D^2(x) + \frac{1}{4}D^4 x - \dots\right]$$

$$= \frac{x}{2}.$$

2. Solve $x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$ [Jan 2014]

Soln:

Given $xy'' + y' = 0$

$\times$ by x on both sides we get

$$x^2y'' + xy' = 0$$

Put $x = e^z \Rightarrow \log x = z$ and $xD = \theta$, $x^2D^2 = \theta(\theta-1)$

and $\theta = \frac{d}{dz}$ then the transformed eqn is

$$[\theta(\theta-1)+\theta]y = 0$$

$$\theta^2 y = 0$$

The A.E is $m^2 = 0 \Rightarrow m = 0, 0$

The C.F is $y = (A+Bz)e^{0z}$

Replace $z = \log x$ and $e^z = x$

$$y = (A + B\log x).$$

3. Find P.I of $(D^2-4D+13)y = e^{2x}$. [April/May 2014]

Soln:

$$P.I = \frac{1}{D^2-4D+13} e^{2x} \quad \text{Replace } D=2$$

$$= \frac{1}{4-8+13} e^{2x} = \frac{e^{2x}}{9}.$$

4. Find the complementary function of $(D^2-2D+1)y = x$.
[April/May 2014]

Soln:

Given $(D^2-2D+1)y = x$

The A.E is $m^2-2m+1 = 0$

$(m-1)^2 = 0 \Rightarrow m = 1, 1.$

$$C.F = (A+Bx)e^x.$$

5. Solve $(D^3-12D+16)y = 0$ (Jan 2015)

Soln:

The A.E is $m^3-12m+16 = 0$.

where $m=2$ is a factor then by using synthetic division method we have $m^2+2m-8=0$.

That is $(m-2)(m+4)=0$

$\therefore$ The roots are $m=2,2,-4$.

Hence the solution is $y=(A+Bx)e^{2x}+Ce^{-4x}$.

6. Find the particular integral of $(D^2+1)y=\sin x$. [Jan 2015]

Soln:

$$P.I = \frac{1}{D^2+1}\sin x \qquad \text{Replace } D^2=-1^2=-1$$

$$= \frac{1}{-1+1}\sin x = \frac{1}{0}\sin x$$

$$= x\frac{1}{2D}\sin x = \frac{-x\cos x}{2}.$$

7. Solve $\frac{d^3y}{dx^3}+3\frac{d^2y}{dx^2}+3\frac{dy}{dx}+2y=0$ (May 2015)

Soln:

Given $(D^3+3D^2+3D+2)y=x$.

The A.E is $m^3+3m^2+3m+2=0$

By using synthetic division method we have

$(m+2)(m^2+m+1)=0$

$m=-2,\ \frac{-1}{2}\pm i\frac{\sqrt{3}}{2}$.

$$C.F = Ae^{-2x}+e^{-x/2}\left(B\cos\frac{\sqrt{3}}{2}x+C\sin\frac{\sqrt{3}}{2}x\right)$$

$$y = Ae^{-2x}+e^{-x/2}\left(B\cos\frac{\sqrt{3}}{2}x+C\sin\frac{\sqrt{3}}{2}x\right)$$

8. Find the P.I of $(D^2+4)y = \sin 2x$. (May 2015)

Soln:

$$P.I = \frac{1}{D^2+2^2} \sin 2x.$$

Replace $D^2 = -2^2 = -4$

$$= \frac{\sin 2x}{-4+4} = \frac{\sin 2x}{0}.$$

$$= \frac{x}{2} \frac{1}{D} (\sin 2x)$$

$$= \frac{x}{2} \int \sin 2x \, dx$$

$$P.I = \frac{-x \cos 2x}{4}.$$

9. Solve $(D^2+3D+2)y = 0$. [Jan 2016]

Soln:

Given $(D^2+3D+2)y = 0$

$\therefore$ A.E is $m^2+3m+2 = 0$

$\Rightarrow m = -1, -2.$

$$y = Ae^{-x} + Be^{-2x}.$$

10. Eliminate y from the following simultaneous eqns

$\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$ (Jan 2016)

Soln:

Given $Dx + y = \sin t \quad \ldots (1)$

$Dy + x = \cos t \quad \ldots (2)$

Solving y from the above two eqns we get

① × D ⟹ $D^2x + Dy = -\cos t$

② ⟹ $Dy + x = \cos t$

Subtracting the above eqns we get

$(D^2-1)x = 0.$

11. Solve $(D^3 - D^2 - D + 1)y = 0.$ [May 2016]

Soln:

The A.E is $m^3 - m^2 - m + 1 = 0$

By using synthetic division method

$(m-1)(m^2-1) = 0$

$m = 1, 1, -1$

Hence complementary function is,

$y = (A + Bx)e^{x} + Ce^{-x}.$

12. Solve $y' + y = \sin x$ using the method of variation of parameters. [May 2016]

Soln:

Method of variation parameters is applicable only for the second order ODE. Hence the given problem is wrong.

13. Solve $(D^2 - 3D + 2)y = e^{5x}$ [May 2017]

Soln:

Given $(D^2 - 3D + 2)y = e^{5x}$

The A.E is $m^2 - 3m + 2 = 0$

2

$-1 \quad -2$

-3

$(m-1)(m-2) = 0$

$m = 1, 2$

$C.F = Ae^{m_1 x} + Be^{m_2 x}$

$C.F = Ae^{x} + Be^{2x}$

$$P.I = \frac{1}{D^2 - 3D + 2} \cdot e^{5x}$$

$$= \frac{1}{(5)^2 - 3(5) + 2} \cdot e^{5x}$$

Replace D by a

$D = 5$

$$= \frac{1}{25 - 15 + 2} e^{5x}$$

$$= \frac{1}{12} e^{5x} \,//$$

$y = C.F + P.I$

$$y = Ae^{x} + Be^{2x} + \frac{1}{12} e^{5x}$$

14. Define second order differential equation. [May 2017]

Soln:

A second order differential equation is an equation involving the unknown function y, its derivatives y' and y'', and the variable x. We will only consider explicit differential equations of the form, Non linear Equations and linear equations.

15. Solve the equation $(D^2+2D-24)y=0$. [Nov/Dec 2018]

Soln:

Gn $(D^2+2D-24)y=0$

The A.E is $m^2+2m-24=0$

$$m=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$a=1,\ b=2,\ c=-24$

$$m=\frac{-2\pm\sqrt{4-4(1)(-24)}}{2(1)}$$

$$=\frac{-2\pm\sqrt{4+96}}{2}$$

$$=\frac{-2\pm\sqrt{100}}{2}$$

$$=\frac{-2\pm 10}{2}$$

$$=-1\pm 5$$

$$=-1+5,\ -1-5$$

$$m=4,-6$$

$$y=Ae^{4x}+Be^{-6x}$$ //

16. Find the P.I of $(D^2+D+1)y=x^2$

Soln:

$$P.I=\frac{1}{D^2+D+1}x^2$$

$$=\frac{1}{1+(D^2+D)}x^2$$

$$=[1+(D^2+D)]^{-1}x^2$$

$[\because (1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots]$

$= [1 - (D^2+D) + (D^2+D)^2 - \dots] x^2$

$= [1 - D^2 - D + D^4 + 2D^3 + D^2 - \dots] x^2$

$= x^2 - D^2(x^2) - D(x^2) + D^4(x^2) + 2D^3(x^2) + D^2(x^2) - \dots$

$= x^2 - \cancel{2} - 2x + 0 + 0 + \cancel{2}$

$P.I = x^2 - 2x$ //

17. Solve $(D^2+5D+6)y = 0$ [April / May 2018]

<u>Soln:</u>

Gn $(D^2+5D+6)y = 0$

The A.E is $m^2+5m+6 = 0$

$(m+2)(m+3) = 0$

$m = -2, -3$

$C.F = Ae^{-2x} + Be^{-3x}$

$y = Ae^{-2x} + Be^{-3x}$ //

18. Find P.I of $(D^2 - 8D + 9)y = \sin 5x$ [April / may 2018]

<u>soln:</u>

Given $(D^2 - 8D + 9)y = \sin 5x$

The A.E is $m^2 - 8m + 9 = 0$

$\begin{array}{ccc} & 9 & \\ 1 & \wedge & 9 \\ & -8 & \end{array}$ ×

$a=1,\ b=-8,\ c=9$

$$m=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$m=\frac{8\pm\sqrt{64-4(1)9}}{2(1)}$$

$$=\frac{8\pm\sqrt{64-36}}{2}$$

$$=\frac{8\pm\sqrt{28}}{2}$$

$$=\frac{8\pm\sqrt{4\times7}}{2}$$

$$=\frac{8\pm2\sqrt{7}}{2}$$

$$=4\pm\sqrt{7}.$$

$m=4+\sqrt{7},\ 4-\sqrt{7}$

$C.F = Ae^{m_1x}+Be^{m_2x}$

$C.F = Ae^{(4+\sqrt{7})x}+Be^{(4-\sqrt{7})x}$

$$P.I=\frac{1}{D^2-8D+9}\sin 5x$$

Replace $D^2=-a^2$
$=-5^2=-25$

$$=\frac{1}{-25-8D+9}\sin 5x$$

$$=\frac{1}{-16-8D}\sin 5x$$

$$=-\frac{1}{8D+16}\sin 5x$$

$$= -\frac{(8D-16)}{(8D+16)(8D-16)} \sin 5x$$

$$= -\frac{(8D-16)}{(8D)^2-(16)^2} \sin 5x$$

$$= -\frac{(8D-16)}{64D^2-256} \sin 5x$$

Replace $D^2 = -5^2$
$= -25$

$$= -\frac{[8D(\sin 5x) - 16\sin 5x]}{64(-25)-256}$$

$$= -\left(\frac{8(5)\cos 5x - 16\sin 5x}{-1600-256}\right)$$

$$= \cancel{+}\frac{40\cos 5x - 16\sin 5x}{\cancel{-}1856}$$

$$= \frac{\cancel{4}[10\cos 5x - 4\sin 5x]}{\cancel{1856}\ 464}$$

$$= \frac{\cancel{2}[5\cos 5x - 2\sin 5x]}{\cancel{464}}$$

$$P.I = \frac{5\cos 5x - 2\sin 5x}{232}$$ //

$y = C.F + P.I$ //

Part - B (11 marks)

1. Solve $x^2 \frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = \log x$. (Jan 2014)

Soln:

Given $(x^2D^2 - xD + 1)y = \log x$

Put $x = e^z \Rightarrow \log x = z$.

Then $xD = \theta$

$x^2D^2 = \theta(\theta - 1)$ and $\theta = \frac{d}{dz}$

then the transformed eqn is

$[\theta(\theta-1) - \theta + 1]y = z$

$[\theta^2 - 2\theta + 1]y = z$.

A.E is $(m^2 - 2m + 1) = 0$

(ie) $(m-1)^2 = 0$ $\therefore m = 1, 1$

$C.F = (A + Bz)e^z$.

$P.I = \frac{1}{\theta^2 - 2\theta + 1} z = [1 - (2\theta - \theta^2)]^{-1} z$

$= [1 + 2\theta - \theta^2 + \ldots]z$

$= [z + 2\theta z - \theta^2 z + \ldots] = z + 2$

$y = C.F + P.I$

$= (A + Bz)e^z + z + 2$.

Replace $z = \log x$ and $e^z = x$ then the soln becomes,

$$y = (A + B\log x)x + \log x + 2.$$

2. solve $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$ (April/May 2014)

<u>Soln:</u>

Given $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$.

The A.E is, $m^3 - 3m^2 + 4m - 2 = 0$

using synthetic division method, we have

$$(m-1)(m^2 - 2m + 2) = 0$$

$$\Rightarrow m = 1, 1 \pm i,$$

$$C.F = Ae^x + e^x(B\cos x + C\sin x).$$

$$P.I_1 = \frac{1}{D^3 - 3D^2 + 4D - 2} e^x \qquad \text{Replace } D = 1$$

$$= \frac{e^x}{1 - 3 + 4 - 2} = \frac{e^x}{0}$$

$$= \frac{1}{3D^2 - 6D + 4} e^x$$

$$= \frac{1}{3 - 6 + 4} e^x = e^x.$$

$$P.I_2 = \frac{1}{D^3 - 3D^2 + 4D - 2} \cos x$$

Replace $D^2 = -1^2 = -1$

$$= \frac{1}{-D+3+4D-2}\cos x$$

$$= \frac{1}{3D+1}\cos x = \frac{(3D-1)}{(3D+1)(3D-1)}\cos x$$

$$= \frac{(3D-1)}{9D^2-1}\cos x \qquad \text{Replace } D^2 = -1^2 = -1$$

$$= \frac{3D(\cos x) - \cos x}{-9-1}$$

$$= \left[\frac{-3\sin x - \cos x}{-10}\right] = \left[\frac{3\sin x + \cos x}{10}\right]$$

$$y = Ae^x + e^x(B\cos x + C\sin x) + e^x + \left[\frac{3\sin x + \cos x}{10}\right].$$

3. Solve $\frac{dx}{dt} + 2x - 3y = t$, $\frac{dy}{dt} - 3x + 2y = e^{2t}$.

[April/May 14]

Soln: Given eqns are written as,

$$(D+2)x - 3y = t \quad \text{—}(1)$$

$$-3x + (D+2)y = e^{2t} \quad \text{—}(2)$$

To eliminate y from (1) & (2),

$(1) \times (D+2) \Rightarrow (D+2)^2 x - 3(D+2)y = (D+2)t$ —(3)

$(2) \times 3 \Rightarrow -9x + 3(D+2)y = 3e^{2t}$ —(4)

Adding (3) & (4) we get

$$[(D+2)^2-9]x = (D+2)t+3e^t.$$

$$(D^2+4D-5)x = 1+2t+3e^t.$$

A.E is $m^2+4m-5=0$

$$(m+5)(m-1)=0$$

$$\Rightarrow m=1,-5$$

$$C.F = Ae^t+Be^{-5t}.$$

$$P.I_1 = \frac{1}{D^2+4D-5}(1+2t)$$

$$= \frac{1}{-5+4D+D^2}(1+2t)$$

$$= \frac{1}{-5[1-(\frac{4D}{5}+\frac{D^2}{5})}(1+2t)$$

$$= -\frac{1}{5}\left[1-\left(\frac{4D}{5}+\frac{D^2}{5}\right)\right]^{-1}(1+2t)$$

$$= -\frac{1}{5}\left[1+\left(\frac{4D}{5}+\frac{D^2}{5}\right)+\cdots\right](1+2t)$$

$$= -\frac{1}{5}\left[(1+2t)+\frac{4}{5}\right].$$

$$P.I_2 = \frac{1}{D^2+4D-5}3e^{2t}$$

$$= \frac{1}{4+8-5}3e^{2t} = \frac{3e^{2t}}{7}.$$

$\therefore$ The soln of x is,

$$x = Ae^t+Be^{-5t}-\frac{1}{5}\left[(1+2t)+\frac{4}{5}\right)+\frac{3}{7}e^{2t}.$$

$$\therefore \frac{dx}{dt} = Ae^{t} - 5Be^{-5t} - \frac{2}{5} + \frac{6}{7}e^{2t}$$

From eqn ①, we have

$$3y = \frac{dx}{dt} + 2x + t$$

$$= Ae^{t} - 5Be^{-5t} - \frac{2}{5} + \frac{6}{7}e^{2t} + 2x + t$$

$$\Rightarrow y = Ce^{t} + De^{-5t} - \frac{2}{15} + \frac{2}{7}e^{2t} + \frac{1}{3}(2x+t)$$

Hence the solns are

$$x = Ae^{t} + Be^{-5t} - \frac{1}{5}\left[(1+2t) + \frac{4}{5}\right] + \frac{3}{7}e^{2t}$$

$$y = Ce^{t} + De^{-5t} - \frac{2}{15} + \frac{2}{7}e^{2t} + \frac{1}{3}(2x+t)$$

$$\left[C = \frac{A}{3},\ D = -\frac{5}{3}\right]$$

4. Solve by the method of variation of parameters

$\frac{d^2y}{dx^2} + y = x\sin x$. [$\because$ Jan 2015]

Soln:

Given DE is written as $(D^2+1)y = x\sin x$

The A.E is $m^2+1=0 \Rightarrow m = \pm i$

$[\because m = -1 = i^2]$

$$C.F = e^{0x}(c_1\cos x + c_2\sin x)$$

$$= c_1\cos x + c_2\sin x = c_1 f_1 + c_2 f_2$$

where $f_1 = \cos x$ $\quad f_2 = \sin x$ $\quad$ and $X = \sec x$

$f_1' = -\sin x$ $\quad f_2' = \cos x$

$\therefore f_1 f_2' - f_1' f_2 = \cos^2 x + \sin^2 x = 1.$

$P.I = Pf_1 + Qf_2$

$$P = -\int \frac{f_2 X}{f_1 f_2' - f_1' f_2} dx$$

$$= -\int \frac{\sin x \cdot x \sin x}{1} dx \qquad [\because X = x \sin x]$$

$$= -\int x \sin^2 x \, dx = -\int x \left[\frac{1 - \cos 2x}{2}\right] dx$$

$$= -\frac{1}{2}\int x dx + \frac{1}{2}\int x \cos 2x \, dx$$

$$= -\frac{x^2}{4} + \frac{1}{2}\left[\frac{x \sin 2x}{2} + \frac{\cos 2x}{4}\right]$$

$$= -\frac{x^2}{4} + \frac{x \sin 2x}{4} + \frac{\cos 2x}{8}.$$

$$Q = \int \frac{f_1 X}{f_1 f_2' - f_1' f_2} dx$$

$$= \int \frac{\cos x \cdot x \sin x}{1} dx = \frac{1}{2}\int x \sin 2x \, dx$$

$$= \frac{1}{2}\left[-\frac{x \cos 2x}{2} + \frac{\sin 2x}{4}\right]$$

$$= -\frac{x \cos 2x}{4} + \frac{\sin 2x}{8}.$$

$$P.I = Pf_1 + Qf_2$$

$$= \left[-\frac{x^2}{4} + \frac{x\sin 2x}{4} + \frac{\cos 2x}{8}\right]\cos x$$

$$+ \left[-\frac{x\cos 2x}{4} + \frac{\sin 2x}{8}\right]\sin x.$$

$$y = C_1\cos x + C_2\sin x + \left[-\frac{x^2}{4} + \frac{x\sin 2x}{4} + \frac{\cos 2x}{8}\right]\cos x$$

$$+ \left[-\frac{x\cos 2x}{4} + \frac{\sin 2x}{8}\right]\sin x.$$

5. Solve $(D^2 - 3D + 2)y = 2\cos(2x+3) + 2e^x$. (Jan 2016)

Soln:

Given $(D^2 - 3D + 2)y = 2\cos(2x+3) + 2e^x$

The A.E is, $m^2 - 3m + 2 = 0$ $\therefore m = 1, 2.$

$$C.F = Ae^x + Be^{2x}.$$

$$P.I_1 = \frac{2}{D^2 - 3D + 2}\cos(2x+3)$$ Replace $D^2 = -2^2$ we get

$$= \frac{2}{-4 - 3D + 2}\cos(2x+3).$$

$$= -\frac{2}{(3D+2)}\cos(2x+3)$$

$$= -2\frac{(3D-2)}{(3D+2)(3D-2)}\cos(2x+3)$$

$$= -2\frac{(3D-2)}{9D^2 - 4}\cos(2x+3)$$ Replace $D^2 = -2^2 = -4$

$$= -2\left[\frac{3D\cos(2x+3) - 2\cos(2x+3)}{-36-4}\right]$$

$$= -2\left[\frac{-6\sin(2x+3) - 2\cos(2x+3)}{-40}\right]$$

$$= 4\left[\frac{3\sin(2x+3) + \cos(2x+3)}{-40}\right]$$

$$= -\left[\frac{3\sin(2x+3) + \cos(2x+3)}{10}\right]$$

$P.I_2 = \dfrac{1}{D^2-3D+2}\, 2e^x$ Replace $D = -1$ we get

$$= 2\,\frac{e^{2x}}{4-6+2} = \frac{2e^{2x}}{0}$$

$= \dfrac{2x}{2D-3}\, e^x$ Replace $D = -1$ we get

$$= \frac{2xe^x}{2-3} = -2xe^x.$$

$$y = C.F + P.I_1 + P.I_2$$

$$= Ae^x + Be^{2x} - \left[\frac{3\sin(2x+3) + \cos(2x+3)}{10}\right] - 2xe^x.$$

6. solve $(D^2-4D+4)y = \cosh 2x + 3$ [May 2016]

Soln:

Given $(D^2-4D+4)y = \cosh 2x + 3 = \frac{1}{2}[e^{2x}+e^{-2x}]+3$

$= \frac{1}{2}e^{2x} + \frac{1}{2}e^{-2x} + 3.$

$\therefore$ The A.E is $m^2-4m+4=0$

$(m-2)^2=0 \Rightarrow m=2,2$

$C.F = (A+Bx)e^{2x}.$

$P.I_1 = \frac{1}{2} \frac{1}{D^2-4D+4} e^{2x}$

Replace $D=2$

$= \frac{1}{2} \frac{1}{4-8+4} e^{2x}$

$= \frac{e^{2x}}{0}$

$= \frac{1}{2} \frac{x}{2D-4} e^{2x}$ Replace $D=2$

$= \frac{1}{2} \frac{x}{4-4} e^{2x} = \frac{1}{2} \frac{xe^{2x}}{0}$

$= \frac{1}{2} \frac{x^2e^{2x}}{2} = \frac{x^2e^{2x}}{4}.$

$P.I_2 = \frac{1}{2} \frac{1}{D^2-4D+4} e^{-2x}$ Replace $D=-2$

$= \frac{1}{2} \frac{1}{4+8+4} e^{-2x} = \frac{e^{-2x}}{32}.$

$P.I_3 = \frac{1}{D^2-4D+4} 3.e^{0x}$ Replace $D=0$

$= \frac{3}{0+4} e^{0x} = \frac{3}{4}.$

$y = C.F + P.I_1 + P.I_2 + P.I_3$

$$= (A+Bx)e^{2x} + \frac{x^2 e^{2x}}{4} - \frac{e^{-2x}}{32} + \frac{3}{4}.$$

7. Solve $\frac{dx}{dt} - y = t$, $\frac{dy}{dt} + x = 1$. [May 2016]

Soln:

Given eqns are

$Dx - y = t$ — (1)

$x + Dy = 1$ — (2)

To eliminate x from (1) & (2).

(1) $\Rightarrow Dx - y = t$ — (3)

(2) $\times D \Rightarrow Dx + D^2y = 0$ — (4)

Sub (3) & (4) we get $(D^2+1)y = -t$

The A.E is $m^2+1=0$

$\Rightarrow m^2 = i^2$ $\therefore m = \pm i$

C.F is $y = A\cos t + B\sin t$

$$P.I = \frac{1}{D^2+1}(-t)$$

$$= -[1+D^2]^{-1} t$$

$$= -[1 - D^2 + D^4 - D^6 + \dots]t$$

$$= -t$$

$\therefore$ The soln is $y = A\cos t + \sin t - t$

Differentiating w.r.to t, $\frac{dy}{dt} = -A\sin t + B\cos t - 1$

$$② \Rightarrow x = 1 - \frac{dy}{dt}$$

$$= 1 - [-A\sin t + B\cos t - 1]$$

$$= 1 + A\sin t - B\cos t + 1$$

$$\therefore x = C\sin t + D\cos t + 2.$$

where $C = A$ & $D = -B$.

The solutions are

$$y = A\cos t + \sin t - t \text{ and } x = C\sin t + D\cos t + 2.$$

7. Using Method of variation of parameters, solve $\frac{d^2y}{dx^2} + y = \sec x$. [Jan 2014]

Soln:

Given $(D^2+1)y = \sec x$

The A.E is $m^2+1=0 \Rightarrow m = \pm i$

$$C.F = e^{0x}(c_1\cos x + c_2\sin x)$$

$$= c_1\cos x + c_2\sin x$$

$$= c_1 f_1 + c_2 f_2$$

where $f_1 = \cos x$ $\quad f_2 = \sin x$ $\quad$ and $X = \sec x$

$f_1' = -\sin x$ $\quad f_2' = \cos x$

$$\therefore f_1 f_2' - f_1' f_2 = \cos^2 x + \sin^2 x = 1.$$

(6)

$P.I = Pf_1 + Qf_2$

$$P = -\int \frac{f_2 X}{f_1 f_2' - f_1' f_2} dx$$

$$= -\int \frac{\sin x \cdot \sec x}{1} dx \qquad [\because X = \sec x]$$

$$= -\int \tan x \, dx \qquad \left[\because \sec x = \frac{1}{\cos x}\right]$$

$$= \log(\cos x)$$

$$Q = \int \frac{f_1 X}{f_1 f_2' - f_1' f_2} dx$$

$$= \int \frac{\cos x \cdot \sec x}{1} dx$$

$$= \int dx = x.$$

$\therefore P.I = Pf_1 + Qf_2$

$= \cos x \log(\cos x) + x \sin x$

$$y = c_1 \cos x + c_2 \sin x + \cos x \log(\cos x) + x \sin x.$$

8. solve the following simultaneous eqns,

$\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$. Given that $x = 2, y = 0$ when $t = 0$.

[Jan 14, Jan.'15, May '15]

Nov/Dec 2017

Soln:

Given $Dx + y = \sin t$ — ①

$Dy + x = \cos t$ — ②

Solving y from the above eqns,

① × D ⇒ $D^2x + Dy = D(\sin t)$

② ⇒ $Dy - x = \cos t$

Subtracting

$(D^2 - 1)x = D(\sin t) - \cos t$

$= \cos t - \cos t$

$\therefore (D^2-1)x = 0$

The A.E is $m^2 - 1 = 0 \Rightarrow m = \pm 1$

C.F ⇒ $x = Ae^{t} + Be^{-t}$

Diff x w.r.to t we get, $\frac{dx}{dt} = Ae^{t} - Be^{-t}$

From ① we have

$y = -\frac{dx}{dt} + \sin t$

$y = -Ae^{t} + Be^{-t} + \sin t.$

The solns are $x = Ae^{t} + Be^{-t}$ and $y = -Ae^{t} + Be^{-t} + \sin t.$

Now applying given initial conditions

(ii) $x=2,\ y=0$ when $t=0$ in the solns of

x and y we get $2=A+B$ and $0=-A+B$.

Solving these two eqns we get $A=B=1$.

Sub A & B values in soln of x and y we get

$$x = e^{t} + e^{-t} = 2\cosh t \quad \text{and}$$

$$y = -e^{t} + e^{-t} + \sin t$$

$$= -2\sinh t + \sin t.$$

9. Solve $(D^3 - 7D - 6)y = (1+x)e^{2x}$. [May 2015]

<u>soln:</u>

The A.E is $m^3 - 7m - 6 = 0$

By synthetic division method the roots are

$m = -1, -2, -3$

$$C.F = Ae^{-x} + Be^{-2x} + Ce^{-3x}.$$

$$P.I = \frac{1}{D^3 - 7D - 6} e^{2x}(1+x)$$

Replace D by $D+2$

$$= e^{2x} \frac{1}{(D^3 + 6D^2 + 12D + 8) - 7(D+2) - 6}(1+x)$$

$$= e^{2x} \frac{1}{-12+5D+6D^2+D^3} (1+x)$$

$$= -\frac{e^{2x}}{12}\left[1-\left(\frac{5D+6D^2+D^3}{12}\right)\right]^{-1} (1+x)$$

$$\left[(1-x)^{-1} = 1+x+x^2+x^3+\cdots\right]$$

$$= \frac{-e^{2x}}{12}\left[1+\left(\frac{5D+6D^2+D^3}{12}\right)+\cdots\right](1+x)$$

$$= \frac{-e^{2x}}{12}\left[(1+x)+\frac{5D}{12}(1+x)+\frac{6D^2}{12}(1+x)+\frac{D^3}{12}(1+x)+\cdots\right]$$

$$= \frac{-e^{2x}}{12}\left[(1+x)+\frac{5}{12}(0+1)+\frac{6}{12}(0+0)+0\right]$$

$$= \frac{-e^{2x}}{12}\left[(1+x)+\frac{5}{12}+0\right]$$

$$y = C.F + P.I$$

$$y = Ae^{-x}+Be^{-2x}+Ce^{-3x}-\frac{e^{2x}}{12}\left[(x+1)+\frac{5}{12}\right]$$

Solve $(D^2+a^2)y = \tan ax$ by method variation of Parameters. [Jan 2016]

Soln:

Given $(D^2+a^2)y = \tan ax$

The A.E is $m^2+a^2=0 \Rightarrow m = \pm ai$.

$$C.F = c_1 \cos ax + c_2 \sin ax$$

$$= c_1 f_1 + c_2 f_2$$

where $f_1 = \cos ax$ $\qquad f_2 = \sin ax$

$f_1' = -a \sin ax$ $\qquad f_2' = a \cos ax$

$$\therefore f_1 f_2' - f_1' f_2 = a\cos^2 ax + a \sin^2 ax$$

$$= a[\cos^2 ax + \sin^2 ax] = a$$

$$P.I = Pf_1 + Qf_2$$

$$P = -\int \frac{f_2 X}{f_1 f_2' - f_1' f_2}\, dx$$

$$= -\int \frac{\sin ax \cdot \tan ax}{a}\, dx$$

$$= -\frac{1}{a}\int \frac{\sin^2 ax}{\cos ax}\, dx = -\frac{1}{a}\int \frac{1-\cos^2 ax}{\cos ax}\, dx$$

$$= -\frac{1}{a}\left[\int \sec ax\, dx - \int \cos ax\, dx\right]$$

$$= -\frac{1}{a}\left[\frac{1}{a}\log(\sec ax + \tan ax] - \frac{\sin ax}{a}\right]$$

$$Q = \int \frac{f_1 X}{f_1 f_2' - f_1' f_2}\, dx$$

$$= \int \frac{\cos ax \cdot \tan ax}{a}\, dx$$

$$= \frac{1}{a}\int \sin ax\, dx = \frac{-\cos ax}{a^2}.$$

$P.I = Pf_1 + Qf_2$

$= -\frac{1}{a}\left[\frac{1}{a}\log(\sec ax + \tan ax) - \frac{\sin ax}{a}\right]\cos ax$

$- \frac{\cos ax}{a^2} \sin ax$

$= -\frac{1}{a^2}\cos ax \log(\sec ax + \tan ax).$

$y = c_1 \cos ax + c_2 \sin ax - \frac{1}{a^2}\cos ax \log(\sec ax + \tan ax)$

11. Using method of variation of parameters, solve $\frac{d^2y}{dx^2} + n^2 y = \sec nx.$ [Nov/Dec 2018]

Soln:

Given $D^2y + n^2y = \sec nx.$

$(D^2+n^2)y = \sec nx$

The A.E is

$m^2+n^2=0 \Rightarrow m^2 = -n^2$

$m = \sqrt{-n^2} \Rightarrow m = \pm ni$

$C.F = e^{0x}(c_1 \cos nx + c_2 \sin nx)$

$C.F = c_1 \cos nx + c_2 \sin nx$

where $f_1 = \cos nx$ $\quad f_2 = \sin nx$

$f_1' = -\sin nx . n$ $\quad f_2' = n.\cos nx$

$= -n \sin nx$

$\therefore f_1 f_2' - f_1' f_2 = \cos nx . n\cos nx - (-n\sin nx)\sin nx$

$= n\cos^2 nx + n\sin^2 nx = n(\cos^2 nx + \sin^2 nx)$

$= n$

$P.I = Pf_1 + Qf_2$

$[\because X = \sec nx]$

$$P = -\int \frac{f_2 X}{f_1 f_2' - f_1' f_2} \, dx$$

$$= -\int \frac{\sin nx \cdot \sec nx}{n} \, dx$$

$$= -\int \frac{\sin nx \cdot \frac{1}{\cos nx}}{n} \, dx$$

$\left[\because \int \tan ax \, dx = \frac{1}{a} \log (\sec ax)\right]$

$$= -\frac{1}{n} \int \tan nx \, dx$$

$$= -\frac{1}{n} \left(\frac{1}{n} \log \sec nx\right)$$

$$= -\frac{1}{n^2} \log (\sec nx)$$

$$= -\frac{1}{n^2} \log \left(\frac{1}{\cos nx}\right)$$

$$= -\frac{1}{n^2} (\log 1 - \log \cos nx)$$

$$= -\frac{1}{n^2} [0 - \log \cos nx]$$

$$= \frac{1}{n^2} \log \cos nx.$$

$$Q = \int \frac{f_1 X}{f_1 f_2' - f_1' f_2} \, dx$$

$$= \int \frac{\cos nx \cdot \sec nx}{n} \, dx$$

$$= \int \frac{\cos nx \cdot \frac{1}{\cos nx}}{n} dx$$

$$= \frac{1}{n} \int dx = \frac{1}{n} x$$

$$P.I = Pf_1 + Qf_2$$

$$= \frac{1}{n^2} \log(\cos nx) \cdot \cos nx + \frac{1}{n} x \cdot \sin nx$$

$$y = C.F + P.I$$

$$= C_1 \cos nx + C_2 \sin nx + \frac{1}{n^2} \log(\cos nx) \cdot \cos nx$$
$$+ \frac{1}{n} x \cdot \sin nx$$ //

solve $(D^2+1)y = x^2 e^{2x} + x \cos x$ [Nov/Dec 2018]

<u>Soln:</u>

Given $(D^2+1)y = 0$

The A.E is $m^2+1 = 0$

$m^2 = -1 \Rightarrow m = \pm i$

$C.F = e^{0x}(A\cos x + B\sin x)$

$C.F = A\cos x + B\sin x$

$$P.I_1 = \frac{1}{D^2+1} \cdot e^{2x} \cdot x^2$$

Replace $D = D+a$
$D = D+2$

$$= e^{2x} \frac{1}{(D+2)^2+1} x^2$$

$$= e^{2x} \frac{1}{D^2+4D+4+1} x^2$$

$$= e^{2x} \frac{1}{D^2+4D+5} x^2$$

$$= e^{2x} \frac{1}{5\left(1+\left(\frac{D^2+4D}{5}\right)\right)} x^2 \quad \left[\because (1+x)^{-1} = 1-x+x^2-x^3+\cdots\right]$$

$$= e^{2x} \frac{1}{5} \left[1+\left(\frac{D^2+4D}{5}\right)\right]^{-1} x^2$$

$$= e^{2x} \frac{1}{5} \left(1-\left(\frac{D^2+4D}{5}\right) + \left(\frac{D^2+4D}{5}\right)^2 - \cdots\right] x^2$$

$$= e^{2x} \frac{1}{5} \left[1 - \frac{D^2}{5} - \frac{4D}{5} + \frac{D^4}{25} + \frac{8D^3}{25} + \frac{16D^2}{25} - \cdots\right] x^2$$

$$= e^{2x} \frac{1}{5} \left(x^2 - \frac{D^2}{5}(x^2) - \frac{4D}{5}(x^2) + \frac{D^4}{25}(x^2)\right.$$

$$\left. + \frac{8D^3}{25}(x^2) + \frac{16D^2}{25}(x^2) - \cdots\right]$$

$$= e^{2x} \frac{1}{5} \left[x^2 - \frac{2}{5} - \frac{4}{5}(2x) + 0 + 0 + \frac{16}{25}(2)\right]$$

$$= e^{2x} \frac{1}{5} \left[x^2 - \frac{8x}{5} + \frac{32}{25} - \frac{2}{5}\right]$$

$$= e^{2x} \frac{1}{5} \left[x^2 - \frac{8x}{5} + \frac{32-10}{25}\right]$$

$$= \frac{e^{2x}}{5} \left[x^2 - \frac{8x}{5} + \frac{22}{25}\right]$$

(11)

$$P.I_2 = \frac{1}{D^2+1} \cdot x\cos x$$

$$= R.P. \text{ of } \frac{1}{D^2+1} \cdot e^{ix} \cdot \cos x$$

$\left[\because e^{i\theta} = \cos\theta + i\sin\theta \right]$
$R.P. \text{ of } e^{i\theta} = \cos\theta$
$R.P. \text{ of } e^{ix} = \cos x$

$$= R.P. \text{ of } e^{ix} \frac{1}{(D+i)^2+1} \cos x$$

Replace $D = D+a$
$D = D+i$

$$= R.P. \text{ of } e^{ix} \frac{1}{D^2+2Di+\cancel{i^2}+\cancel{1}} \cos x$$

$$= R.P. \text{ of } e^{ix} \frac{1}{D^2+2Di} \cos x$$

$$= R.P. \text{ of } e^{ix} \frac{1}{-1+2Di} \cos x$$

Replace $D^2 = -a^2$
$= -(1^2)$

$$= R.P. \text{ of } e^{ix} \frac{1}{-[1-2Di]} \cos x$$

$$= -R.P. \text{ of } e^{ix} \frac{[1+2Di]}{(1-2Di)(1+2Di)} \cdot \cos x$$

$$= -R.P \text{ of } e^{ix} \frac{(1+2Di)}{1^2-(2Di)^2} \cos x$$

$$= -R.P \text{ of } e^{ix} \frac{[\cos x + 2Di(\cos x)]}{1-4D^2i^2}$$

Replace $D^2 = -(1^2)$

$$= -R.P \text{ of } e^{ix} \frac{(\cos x + 2Di(\cos x)]}{1-4(-1)(-1)}$$

$$= \cancel{-} R.P \text{ of } e^{ix} \frac{[\cos x + 2i(-\sin x)]}{\cancel{-}3}$$

$$= R.P \text{ of } (\cos x + i\sin x)[\cos x - 2i(+\sin x)]$$

$= R.P \text{ of } [\cos^2 x - 2i\sin x\cos x + i\sin x\cos x$
$- 2i^2 \sin^2 x]$ $(\because i^2 = -1)$

$= R.P \text{ of } [\cos^2 x - 2i\sin x\cos x + i\sin x\cos x + 2\sin^2 x]$

$= \cos^2 x + 2\sin^2 x$

$\therefore y = C.F + P.I_1 + P.I_2$

$$y = A\cos x + B\sin x + \frac{e^{2x}}{5}\left(x^2 - \frac{8x}{5} + \frac{22}{25}\right)$$

$$+ \cos^2 x + 2\sin^2 x$$

13. Using method of variation of parameters, solve

$\frac{d^2y}{dx^2} + y = \operatorname{cosec} x.$ [Nov / Dec 2017]
May 2017

Soln:

Given $(D^2+1)y = \operatorname{cosec} x.$

The A.E is $m^2+1=0.$ $\therefore m = \pm i$

$C.F = e^{0x}(c_1\cos x + c_2\sin x)$

$= c_1\cos x + c_2\sin x = c_1 f_1 + c_2 f_2$

where $f_1 = \cos x$ $f_2 = \sin x$ $X = \operatorname{cosec} x.$

$f_1' = -\sin x$ $f_2' = \cos x.$

$\therefore f_1 f_2' - f_1' f_2 = \cos^2 x + \sin^2 x = 1.$

$P.I = Pf_1 + Qf_2$

$$P = -\int \frac{f_2 X}{f_1 f_2' - f_1' f_2}$$

$$P = -\int \frac{f_2 X}{f_1 f_2' - f_1' f_2} dx$$

$$= -\int \frac{\sin x \cdot \operatorname{cosec} x}{1} dx \qquad \left[\because \operatorname{cosec} x = \frac{1}{\sin x}\right]$$

$$= -\int dx = -x$$

$$Q = \int \frac{f_1 X}{f_1 f_2' - f_1' f_2} dx \qquad [\because X = \operatorname{cosec} x]$$

$$= \int \frac{\cos x \cdot \operatorname{cosec} x}{1} dx$$

$$= \int \cot x \, dx = \log \sin x.$$

$P.I = Pf_1 + Qf_2 = -x\cos x + (\log \sin x) \sin x.$

$y = c_1 \cos x + c_2 \sin x - x\cos x + \sin x \log x$,,

14. Solve $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = \sin(\log x)$

[May 2017, Jan 2016]

Soln:

Given $x^2D^2y + 4xDy + 2y = \sin(\log x)$

$$(x^2D^2 + 4xD + 2)y = \sin(\log x)$$

Put $x = e^z$, $\log x = z$

Sub $xD = \theta$, $x^2D^2 = \theta(\theta-1)$ where $\theta = \frac{d}{dz}$

$$(\theta(\theta-1) + 4\theta + 2)y = \sin z$$

$$(\theta^2 - \theta + 4\theta + 2)y = \sin z$$

$$(\theta^2 + 3\theta + 2)y = \sin z$$

The A.E is $m^2 + 3m + 2 = 0$

2

1 2

3

$$(m+1)(m+2) = 0$$

$$m = -1, -2$$

$$C.F = Ae^{-z} + Be^{-2z}$$

$$P.I = \frac{1}{\theta^2 + 3\theta + 2} \cdot \sin z$$

Replace $\theta^2 = -a^2$
$= -(1^2)$

$$= \frac{1}{-1 + 3\theta + 2} \cdot \sin z$$

$$= \frac{1}{3\theta + 1} \sin z = \frac{1}{(3\theta+1)} \frac{(3\theta-1)}{(3\theta-1)} \sin z$$

$$= \frac{(3\theta-1)\sin z}{(3\theta)^2 - 1^2}$$

$$= \frac{3\theta \sin z - \sin z}{9\theta^2 - 1}$$

Replace $\theta^2 = -1^2$

$$= \frac{3\theta \sin z - \sin z}{9(-1) - 1}$$

$$= \frac{3\cos z - \sin z}{-10}$$

$y = C.F + P.I$

$$y = Ae^{-z} + Be^{-2z} + \frac{3\cos z - \sin z}{-10}$$

Replace $x = e^z$, $\log x = z$

$$y = Ax^{-1} + Bx^{-2} + \frac{3\cos(\log x) - \sin(\log x)}{-10}$$ //

15. Solve $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{2x} + x^2 + x$. [May 2019]

Soln:

Given $D^3y + 2D^2y + Dy = e^{2x} + (x^2 + x)$

$(D^3 + 2D^2 + D)y = e^{2x} + x^2 + x$

The A.E is $m^3 + 2m^2 + m = 0$

$m(m^2 + 2m + 1) = 0$

$m = 0$, $m^2 + 2m + 1 = 0$

$(m+1)(m+1) = 0$

$m = -1, -1$

$\therefore m = 0, -1, -1$

C.F $= (Ax+B)e^{mx} + Ce^{m_3 x}$ $[\because e^0 = 1]$

$= (Ax+B)e^{-x} + Ce^{0x}$

C.F $= (Ax+B)e^{-x} + C$ //

$$P.I_1 = \frac{1}{D^3+2D^2+D} \cdot e^{2x}$$

$$= \frac{1}{2^3+2(2)^2+2} e^{2x} \qquad \text{Replace } D = a, \; D = 2$$

$$= \frac{e^{2x}}{8+8+2} = \frac{e^{2x}}{18} \;//$$

$$P.I_2 = \frac{1}{D^3+2D^2+D} \cdot (x^2+x)$$

$$= \frac{1}{D\left[1+\frac{D^3+2D^2}{D}\right]} (x^2+x)$$

$$= \frac{1}{D\left[1+\left(\frac{D^3}{D}+\frac{2D^2}{D}\right)\right]} (x^2+x)$$

$$= \frac{1}{D\left[1+(D^2+2D)\right]} (x^2+x)$$

$$= \frac{1}{D} \left[1+(D^2+2D)\right] (x^2+x)$$

$$(1+x)^{-1} = 1-x+x^2-x^3+\cdots$$

$$= \frac{1}{D}\left[1-(D^2+2D)+(D^2+2D)^2-\cdots\right](x^2+x)$$

$$= \frac{1}{D}\left[1-D^2-2D+D^4+4D^3+4D^2-\cdots\right](x^2+x)$$

$$= \left(\frac{1}{D} - \frac{D^{\cancel{2}}}{\cancel{D}} - \frac{2\cancel{D}}{\cancel{D}} + \frac{D^{\cancel{4}\,3}}{\cancel{D}} + 4\frac{D^{\cancel{3}\,2}}{\cancel{D}} + \frac{4D^{\cancel{2}\,1}}{\cancel{D}} - \cdots\right](x^2+x)$$

$$= \left[\frac{1}{D}(x^2+x) - D(x^2+x) - 2(x^2+x) + D^3(x^2+x)\right.$$

$$\left. + 4D^2(x^2+x) + 4D(x^2+x) - \cdots\right]$$

$$= \frac{x^3}{3} + \frac{x^2}{2} - 2x - 1 - 2x^2 - 2x + 0 + 0 + 4(2) + 4(0)$$

$$+ 4(2x+1)$$

$$= \frac{x^3}{3} + \frac{x^2}{2} - 2x^2 - 2x - 2x + 8x - 1 + 8 + 4$$

$$= \frac{x^3}{3} + \frac{x^2 - 4x^2}{2} - 4x + 8x + 11$$

$$= \frac{x^3}{3} - \frac{3x^2}{2} + 4x + 11 \;//$$

$$\therefore y = C.F + P.I_1 + P.I_2$$

$$y = (Ax+B)e^{-x} + C + \frac{e^{2x}}{18} + \frac{x^3}{3} - \frac{3x^2}{2} + 4x + 11 \;//$$